Guided Lab: Creating and Using a Custom Ansible Inventory

Lab Objective

In this lab, you will:

- Create your first custom Ansible inventory file
- Define groups like `[web]` and `[db]` using the `.ini` format
- Associate SSH user and key for all hosts
- Verify connectivity using the `ping` module
- Get hands-on experience with real-world grouping structure

Lab Prerequisites

Before starting this lab, make sure you have:

- A **control node** with Ansible installed (local machine, WSL, or cloud VM)

- At least one **managed node** (Azure VM) with SSH key-based access already set up
- The public IP of your managed node(s)
- Your private key located at `~/.ssh/id_rsa` (or correct path)

> If you haven't completed SSH setup yet, go back to the Lab in Lecture 01 and complete that first.

Step 1: Create Your Lab Folder

Create a new folder to hold your Ansible project and inventory file:

```
mkdir ansible-inventory-lab
cd ansible-inventory-lab
```

Step 2: Create a Static Inventory File (.ini Format)

Let's start with the simple and common `.ini` format.

Create a file called `inventory.ini`:

```
vi inventory.ini
```

Add the following:

```
[web]
<VM_PUBLIC_IP_1>

[db]
<VM_PUBLIC_IP_2>

[all:vars]
ansible_user=azureuser
ansible_ssh_private_key_file=~/.ssh/id_rsa
```

Replace `<VM_PUBLIC_IP_1>` and `<VM_PUBLIC_IP_2>` with your actual VM IP addresses.

> If you have only one VM, put the same IP under both `[web]` and `[db]` just for testing purposes.

Save and exit.

Step 3: Run Your First Ping Test

Let's now test if Ansible can connect to your remote servers using this inventory:

```
ansible all -i inventory.ini -m ping
```

You should see a `SUCCESS` response from each listed IP, like:

```
<IP> | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
```

If it fails:

- Double-check your IP addresses
- Confirm that SSH access is working
- Check if your private key path in the inventory file is correct
- Verify that port 22 is open on your VM's security group

Step 4: Create a Composite Group (Optional)

Let's add a new group that includes both `[web]` and `[db]` :

Update your `inventory.ini` file:

```
[web]
<VM_PUBLIC_IP_1>

[db]
<VM_PUBLIC_IP_2>

[production:children]
web
db

[all:vars]
```

```
ansible_user=azureuser
ansible_ssh_private_key_file=~/.ssh/id_rsa
```

Now try pinging the `production` group only:

```
ansible production -i inventory.ini -m ping
```

You can also target just `web` or `db` if needed:

```
ansible web -i inventory.ini -m shell -a "uptime"
```

Step 5 (Optional): Try YAML Inventory Format

If you’re curious, here’s how to represent the same thing in YAML.

Create a file `inventory.yaml`:

```yaml
all:
  children:
    web:
      hosts:
        web1:
          ansible_host: <VM_PUBLIC_IP_1>
    db:
      hosts:
        db1:
          ansible_host: <VM_PUBLIC_IP_2>
  vars:
    ansible_user: azureuser
    ansible_ssh_private_key_file: ~/.ssh/id_rsa
```

Test with:

```
ansible all -i inventory.yaml -m ping
```

Best Practices Recap

- Always use a custom inventory file per project
- Keep your inventory files version-controlled (e.g., in Git)
- Use groups (`[web]` , `[db]` , `[production:children]`) to simplify targeting
- Store variables like `ansible_user` and `ansible_ssh_private_key_file` under `[all:vars]`
- Prefer `.ini` format for small projects, `.yaml` for more complex ones

Lab Completed

You now know how to:

- Write a static Ansible inventory in `.ini` and `.yaml` formats
- Define hosts, groups, and shared connection variables
- Organize your servers cleanly and test connections using the `ping` module
- Use grouping for scalable and readable infrastructure automation

This inventory structure is the foundation for everything you'll do next with Ansible.

Student Assignment: Build and Test Your Own Inventory

Objective:

Create and test your own custom inventory structure based on real or test IPs.

Tasks:

1. Create a new file called `my-inventory.ini`

2. Add three groups: `[app]` , `[db]` , and `[monitoring]`

3. Use real or duplicate IPs under each group

4. Add a `[production:children]` group that includes all three

5. Add the `[all:vars]` block to define SSH username and private key path

6. Run the following tests:

- `ansible all -i my-inventory.ini -m ping`
- `ansible app -i my-inventory.ini -m shell -a "uptime"`
- `ansible production -i my-inventory.ini -m command -a "whoami"`

7. Submit your `.ini` file and command outputs as part of the assignment (if applicable)